

Danmarks
Tekniske
Universitet



Assignment # 1

Wind Turbine Technology and Aerodynamics

AUTHORS


Szczepan Letkiewicz - s244067

November 18, 2024

Contents

1	Introduction	1
2	Optimal Power coefficient	1
3	Rated Wind Speed and Max RPM	3
4	Pitch Control and Power Limitation	4
5	Aerodynamic Load Comparison: Ashes vs. BEM	6
6	Annual Energy Production and Cut-Off Impact	8
7	Optimizing Chord and Flow Angle for Max C_P	9
	List of Figures	I
	List of Tables	I
	Nomenclature	I
	References	I
8	Appendix A	II
9	Task division	III

1 Introduction

This report outlines the procedure of applying the Blade Element Method (BEM) algorithm to analyze the performance of a DTU 10 MW virtual experimental wind turbine.

The BEM algorithm is utilised to predict key metrics of a wind turbine such as power output, torque, and efficiency, by computing the normal and tangential force, p_n and p_t . This process is illustrated in Figure 1, which showcases the algorithm and how the inputs can be adjusted to compute the exerted forces for various wind conditions or blade settings. For more details on BEM equations and calculations, refer to Hansen [1].

This paper consists of six sections looking to identify and optimize the wind turbine's performance. Starting with the optimal power coefficient, the ideal pitch angle θ_p and tip speed ratio λ are found before investigating the wind turbine's rated wind speed and maximum rotational speed. After this, the blade-preferred arrangement is investigated to achieve static control across the wind turbine range of operating wind speeds. These calculations are first verified by comparing the BEM calculations with a simulation software *Ashes*, before computing the expected annual energy production and suggesting design adjustments to the blade's chord length c and blade angle with the rotor plane θ .

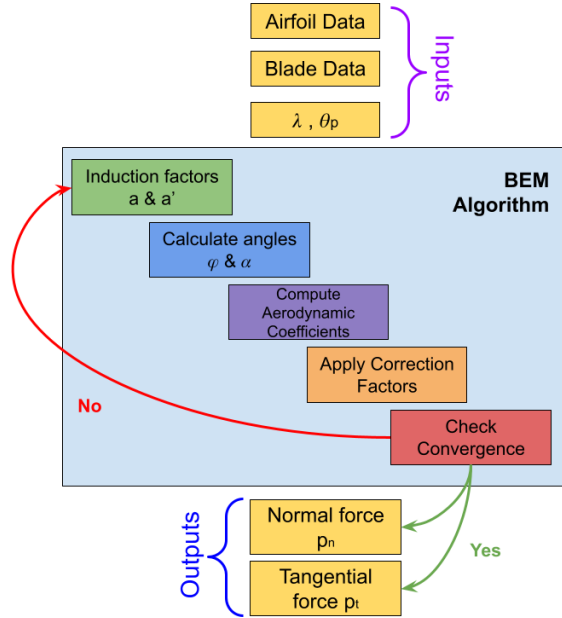


Figure 1: Process diagram of BEM algorithm used to analyze Wind turbine performance.

2 Optimal Power coefficient

The highest power coefficient, $C_{p,max}(\lambda_{max}, \theta_{p,max})$, is obtained by implementing the BEM algorithm for a pitch angle θ_p ranging from -4 and 3 degrees, and tip speed ratio λ between 5 and 10 . The power coefficient C_P is computed using the following equation:

$$C_P = \frac{\omega B \int_0^R r p_t dr}{\frac{1}{2} \rho \pi R^2 V_0^3} \quad (1)$$

and the thrust coefficient C_T is found with:

$$C_T = \frac{B \int_0^R p_n dr}{\frac{1}{2} \rho \pi R^2 V_0^2} \quad (2)$$

The aerodynamic loads p_n and p_t needed to calculate the coefficients are obtained from the BEM algorithm. This is achieved by setting the wind speed V_0 to 8 [m/s] and varying rotational speed ω [rad/s], which produces a range of tip speeds λ using the relation $\lambda = \frac{\omega R}{V_0}$. For this assignment, an iteration step of 0.1 has been chosen for the two values.

Classical Glauert and Madsen correction are methods [1] implemented to find the highest obtainable power coefficient with optimal values of λ and θ_p in the BEM algorithm. The contour plots of the power coefficient together with the corresponding thrust coefficient are presented by the correction method respectively in Figure 2 and Figure 3.

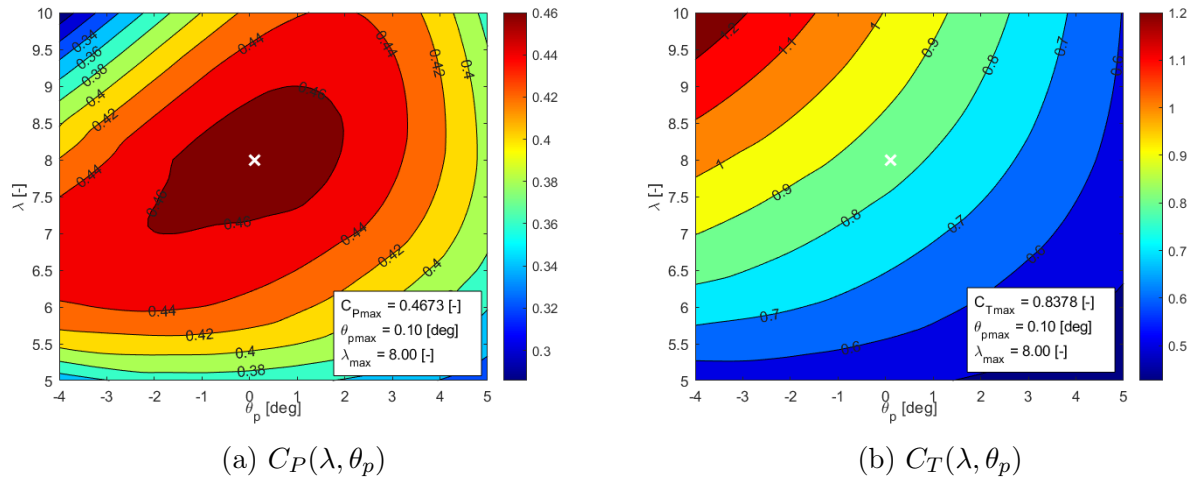


Figure 2: Contour plots using Glauert correction as momentum equation

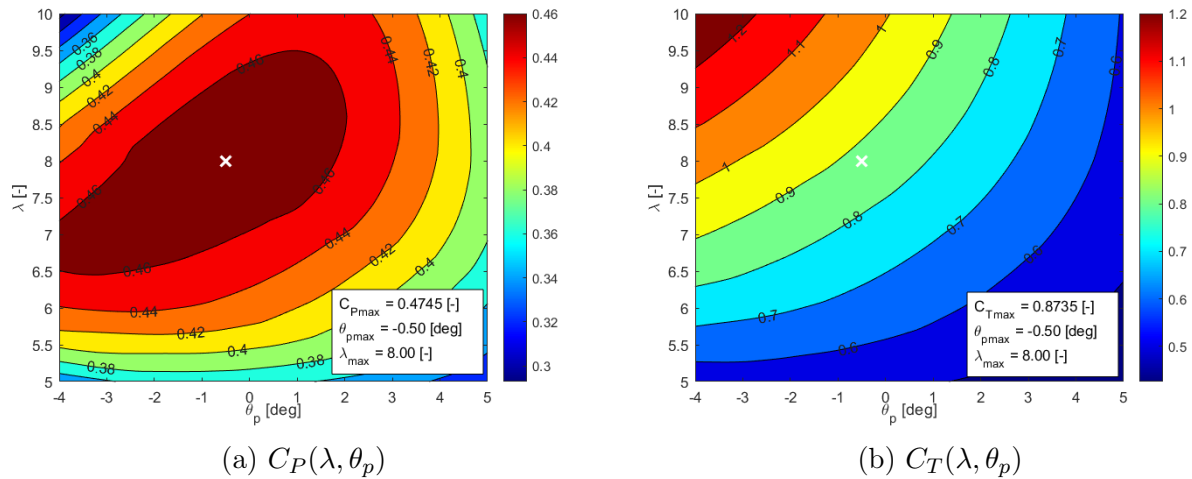


Figure 3: Contour plots using Madsen correction as momentum equation

Based on the contour plots, Madsen predicts a larger C_P value (Table. 1) in comparison to Glauert, by 0.0072. Both values are between 0.46 to 0.48, which is lower than the

maximum theoretical efficiency of $C_P = 16/27 = 0.593$ given by the Betz limit. This limit comes from the conservation of mass and energy and states that some kinetic energy of the wind is lost such that it can flow past the turbine [1].

For both methods the maximum C_P is at a tip speed ratio of 8, however, the methods differ in values of θ_p (Table 1). This result may be due to the difference in dynamics the methods aim to solve. Glauert focuses on where the linear momentum theory breaks down, specifically for high tip speed ratios [1], whereas Madsen focuses on capturing the rotational effects [2].

Along with the highest obtainable C_P , the contour plot's slope shows the sensitivity of the turbine's performance to changes in λ and θ_p . For the Madsen correction plot (Figure 3a) the slope is seen to be much thicker in comparison to Glauert, suggesting that the turbine may still perform close to maximum C_P for different combinations of λ and θ_p . Glauert on the other hand has a much more compressed contour, emphasizing the sensitivity to changes in the variables.

	Glauert	Madsen
$C_{P,max}$	0.4673	0.4745
$\theta_{p,max}$	0.1	-0.5
λ_{max}	8	8
C_T	0.8378	0.8735

Table 1: Highest obtainable power coefficient, and the corresponding tip speed ratio and pitch angle values.

For the remainder of this report, the values obtained from the Glauert correction method have been used, because it is more commonly implemented in software such as Ashes.

3 Rated Wind Speed and Max RPM

The wind turbine is designed to operate between the wind speeds of $V_{in} = 4[\text{m/s}]$ and $V_{out} = 25[\text{m/s}]$ [1]. Therefore the power of the wind turbine for a C_P of 0.4673 is taken from the numerator of equation 1 ($P = \omega B \int_0^R r p_t dr$) and plotted (??) for the range of operational wind speeds V_0 mentioned. The graph shows that the power produced is 0.45 [MW] for the cut-in velocity at which the turbine starts working. Also, the rated power of the turbine P_{rated} of 10.64 [MW] is reached at a wind speed of 11.42 [m/s] denoted as the rated windspeed V_{rated} .

For wind speeds larger than V_{rated} the wind turbine constrains its rotational speed ω to prevent damages in blades and generator from introducing stall conditions. This rotational speed is known as ω_{max} which has a value of 1.02 [rad/s], as indicated in the $\omega(V_0)$ plot (Figure 4). Up to the V_{rated} , the relation is linear based on the tip speed ratio $\lambda = \frac{\omega R}{V_0}$ meaning that the slope of is $\lambda_{max}/R = 0.0897$. For higher wind speeds, the rotational speed is capped by the wind turbine's active control system. According to [3], the DTU 10MW turbine uses

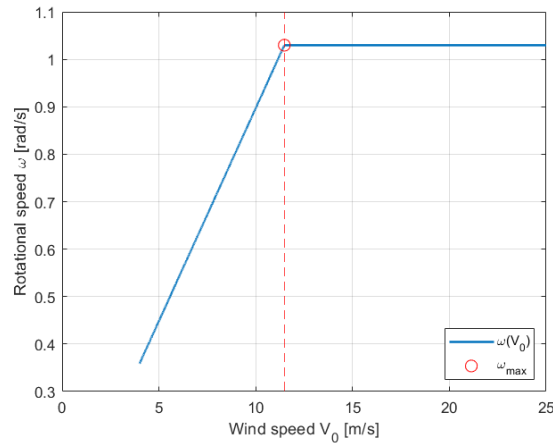


Figure 4: Rotational speed of wind turbine as a function of wind speed, for operational velocities.

Variable-speed pitch regulation (VSPR), which implements both blade pitch regulation as well as electromagnetic generator torque regulation to keep the rotational speed constant. The performance of the controller is beyond the scope of this report, therefore instantaneous control is assumed and the rotational speed is constant instantly after V_{rated} .

4 Pitch Control and Power Limitation

During operation of the wind turbine, the collective pitch angle is actively controlled by the VSPR controller to limit the loads experienced by a turbine and, consequently, keep turbine rotational speed (ω_{max}) and power (P_{rated}) at a constant level. Power is kept constant by increasing the pitch angle which causes the blade sections to experience a lower angle of attack, through the relation $\alpha = \phi - (\theta_p - \beta)$, which leads to lower thrust forces.

Figure 5 presents blade pitch angle as a function of wind speed. It should be noted that the angle is analyzed only for the operational range of wind speed. Initially, the blade pitch angle is kept constant at the value corresponding to the maximum power coefficient, as discussed in section 2. Once the wind speed exceeds V_{rated} , the pitch angle increases. The computed results are validated with data available for stiff structure in DTU 10MW WT report [3]. It can be concluded that data from both sources is almost identical. The only big disparity is associated with low speeds when the pitch controller is activated for low wind speeds (in DTU 10MW report [3]) which is beyond the scope of this project.

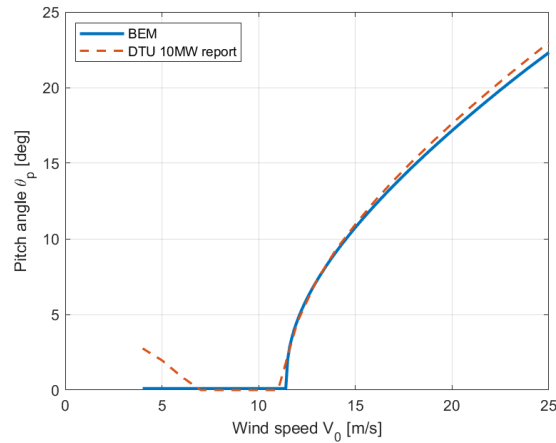


Figure 5: Blade pitch angle as a function of wind speed.

Having determined the blade pitch for each wind speed, characteristic curves for the turbine can be constructed. Figure 6 presents the thrust and power curve for the operational range of wind speeds. Initially, both power and thrust increase until rated windspeed V_{rated} is reached. Then, power is fixed constant, whereas thrust decreases. The highest value of thrust occurs at wind speed equal to V_{rated} . Similarly to the pitch angle curve, the curves align with high-fidelity data curves from DTU 10MW report [3] having marginal discrepancies.

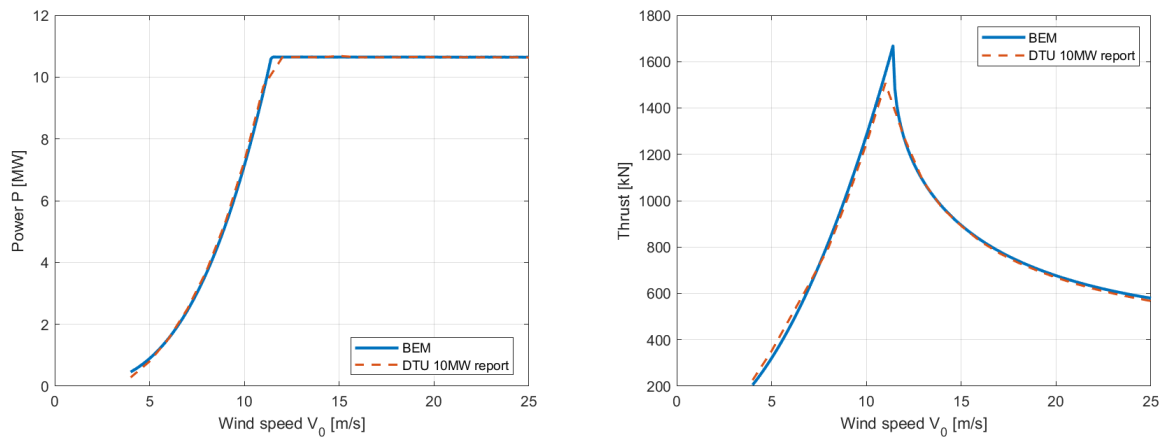


Figure 6: Power and thrust curves as a function of wind speed.

Lastly, non-dimensional coefficients for power and thrust as a function of operational windspeed are plotted in Figure 7. The coefficients express the ratio of obtained power/thrust by the available power/thrust (equations 1,2). Therefore, both coefficients are constant at $C_{P,max} = 0.4673$ and corresponding $C_T = 0.8387$ between cut-in and rated wind speeds. For greater wind speeds, both coefficients decrease since the obtainable maximum power is not utilised, by the controller, to prevent damage to the turbine. Comparison with DTU 10MW

report [3] reveals accurate prediction for both coefficients except small discrepancy occurring for low wind speeds. This is caused by the activation of a pitch controller for low wind speeds and higher fidelity analysis methods which are beyond the scope of this project.

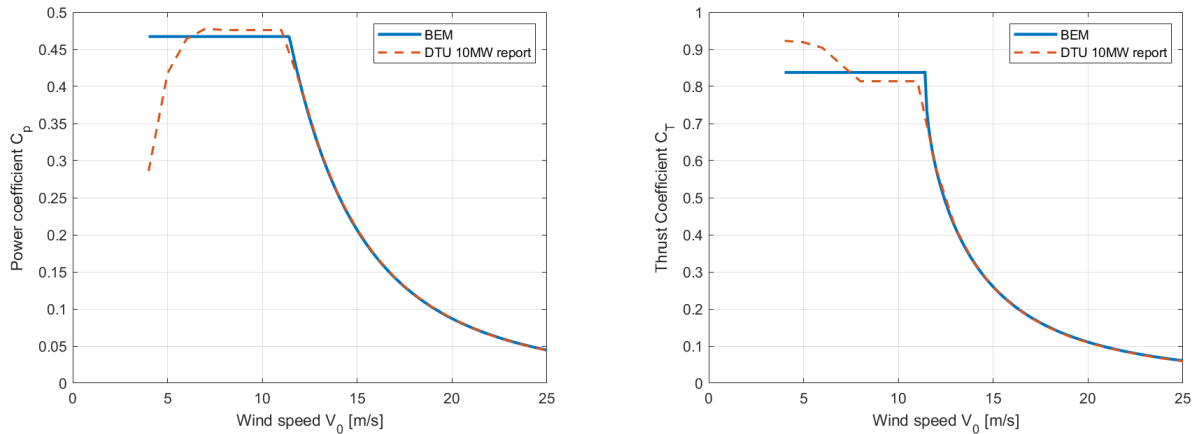


Figure 7: Power and thrust coefficients as a function of wind speed.

5 Aerodynamic Load Comparison: Ashes vs. BEM

To compare the aerodynamic loads produced by the BEM algorithm, identical assumptions must be applied to the *Ashes* program so that the physical conditions are replicated accurately. The following settings have been applied in *Ashes*:

- **Wind setting** → Uniform velocity - A uniform wind speed is assumed across the entire cross-section when applying BEM.
- **Shear stress** → off - The BEM algorithm does not account for shear.
- **Generator** → off - Not account for in BEM
- **Analysis type** → loads only - No structural components are considered in BEM.
- **Structural damping** → off - No structural components are considered in BEM.
- **Tower shadow** → off - The tower is not taken into account in BEM.
- **Glauert correction method** → Empirical - The same as the correction method used in BEM.
- **Hub loss correction factor** → off - Not account for in BEM.
- **Partially submerged** → off - Not account for in BEM.

- **Initial condition for $V_0 = 5, 9 \text{ \& } 11 \rightarrow \lambda = 8 \text{ \& } \theta_p = 0.1$** - Based on findings in Section 3.
- **Initial condition for $V_0 = 20 \rightarrow \lambda = 4.57 \text{ \& } \theta_p = 17.16$** - Based on findings in Section 3.

A comparison of the two methods (Figure 8) indicates that the produced data shows a high degree of similarity and correlation. For wind speeds below V_{rated} , the data is almost a perfect match, and only some discrepancies are noticeable for the wind speed of 20 [m/s]. One clear difference between the two methods is how the root of the blade, which is a cylinder, is accounted for. The symmetry of the cylinder makes it difficult to predict the polarity of the aerodynamic forces, which are negative for BEM and positive for *Ashes*. This numerical discrepancy can also be seen in the tangential force, specifically for 20 [m/s] wind speed.

The normal force P_n , referred to as the thrust force, is seen (Figure 8a) to increase with wind speed until it exerts the highest force for a wind speed of 11 [m/s]. The decrease in normal force seen for a wind speed of 20 [m/s] is due to the blade pitch control discussed in Section 4. The pitch angle is purposely adjusted to limit the turbine's power output to protect its structural integrity. Considering the force distribution along the length of the rotor blade, the highest exerted force for wind speeds below V_{rated} is located near the tip, at a length of approximately $r \approx 80$ [m]. This is due to the tip of the blade moving at a higher linear velocity $V_{tip} = \omega \cdot r$ which contributes to a greater relative wind speed resulting in higher aerodynamic forces. As illustrated in the velocity triangle in [1]. The reason why it is largest at $r = 80$ [m] and not closer to R is because of the tip loss effect which is accounted for by Prandtl's tip loss factor used in BEM. Nevertheless, the wind speed of 20 [m/s] is seen to exhibit its largest force at roughly $r \approx 30$ [m], which stems from the adjusted pitch angle yielding a more optimal angle of attack, and hence aerodynamic performance of the airfoil, in that section.

On the other hand, the tangential force P_t , referred to as the torque, is seen (Figure 8b) to exhibit its largest value for a wind speed of 20 [m/s]. The forces once again differ for wind speeds below V_{rated} , which are observed to exhibit a similar distribution along the length of the rotor. The 20 [m/s] wind speed, however, appears to hold its highest force at $r \approx 30$ [m], presumably following the same reasoning as the normal force at 20 [m/s] wind speeds. It is important to note that the power, which is limited to 10.64 [MW] for wind speeds above V_{rated} is constant. This is reflected by the area under the curve which is perceived to be around the same size for 11[m/s] and 20 [m/s].

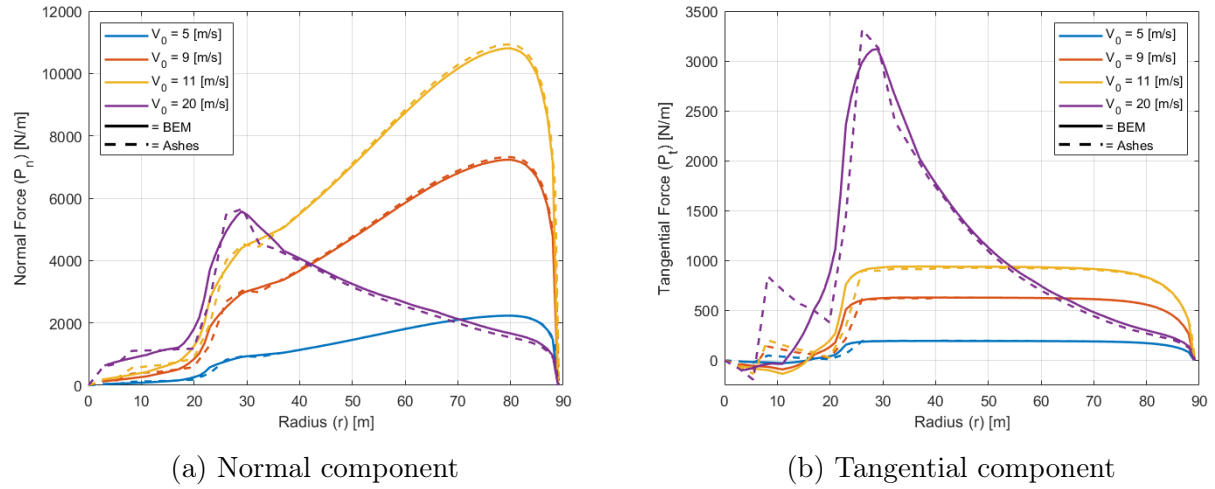


Figure 8: Normal and tangential forces as a function of radius for a range of wind speeds computed using Ashes and BEM algorithm.

6 Annual Energy Production and Cut-Off Impact

Annual Energy Production (AEP) is calculated according to the following formula:

$$\text{AEP} = \sum_{i=1}^{N-1} \frac{1}{2} (P(V_{i+1}) + P(V_i)) \cdot f(V_i < V_o < V_{i+1}) \cdot 8760, \quad (3)$$

where N stands for the number of defined segments in the BEM algorithm, $P(V_i)$ for power produced at the windspeed V_i and $f(V_i < V_o < V_{i+1})$ for Weibull probability density function, described by Hansen [1].

Table 2: Annual Energy Production for cut-out speeds $V_{out} = 25[\text{m/s}]$ and $V_{out} = 20[\text{m/s}]$.

	$V_{out} = 25[\text{m/s}]$	$V_{out} = 20[\text{m/s}]$
AEP [MWh]	38398.35	37510.38

AEP is computed for two scenarios of cut-out speeds $V_{out} = 25[\text{m/s}]$ and $V_{out} = 20[\text{m/s}]$ for identical Weibull parameters being $k = 1.9$ and $A = 9[\text{m/s}]$. The calculations yield results summarized in Table 2. It can be noticed that a decrease in the energy production between both speeds is small amounting to 2.31% of the original value for the $V_{out} = 25[\text{m/s}]$. Opting for a lower cut-out speed has a few advantages.

Firstly, with increasing wind speed, tangential force loading increases and reaches its maximum value closer to the root of the blade. Considering the shape of traditional rotor blades, a thicker cross-section can be seen closer to the root, suggesting that it may be easier to implement structural reinforcements in that area and any potential maintenance.

Secondly, higher loading implies deflections of the blade, specifically near the blade tip. This means that a larger clearance between the blade tip and the turbine tower must be assured. This is especially important since the assumption of stiff blades used in BEM does not hold in reality. Higher, unexpected wind speeds may occur in stormy or harsh weather conditions with frequent turbulent gusts that may cause a local increase of lift leading to structural bending of the blade.

Furthermore, a lower cut-out speed decreases the range of operational pitch angles used by the controller to limit the produced power.

Having all these reasons in mind, one has to perform a trade-off between an increase of turbine AEP and maintenance/manufacturing/development costs associated with turbine operating at higher wind speeds.

7 Optimizing Chord and Flow Angle for Max C_P

Design of the blade section with the selected airfoil is limited to two parameters that can vary - chord c and twist angle β (rotor plane θ is given as $\theta_p - \beta$). Finding the optimum combination of these two parameters allows for maximizing the local power coefficient yielding a more desirable performance of the blade.

This section focuses on a multivariate optimization problem with the optimization function defined as follows:

$$\max_{\substack{0 < c < 3 \\ \theta \in \mathbb{R}}} C_p = \frac{r}{R} \frac{\lambda(1-a)^2 C_t \sigma}{\sin^2 \phi} \quad (4)$$

where λ stand for an optimum tip speed ratio, C_t for tangential load coefficient, a for local induction factor, ϕ for local flow angle, σ for section solidity, r for radial position of the segment and R for rotor radius. While exact relations between variables, requiring the introduction of multiple equations from Blade-Element-Momentum theory, are not of interest, it is important to highlight that solidity σ is a function of c and tangential load coefficient C_t is a function of θ .

The gradient ascent optimization method is chosen to locate the optimum condition. As the problem concerns two variables, partial derivatives for each variable are numerically computed. An initial guess for values of chord and theta (composed of twist and optimum blade pitch angle for $\lambda = 8$) are taken from blade parameters provided, as it is assumed that the blade is already optimized for its performance. This data is summarized in Table 3.

Table 3: Initial values of local power coefficient, chord and angle θ .

	Local power coefficient C_p [-]	Chord c [m]	Initial angle θ [deg]
Initial conditions	0.5279	2.33	-2.180

To ensure convergence of the optimization scheme, different values of step length and

tolerance were tested through trial and error. The step length was determined to be 0.002 and the stopping condition of the absolute difference between two consecutive iterations of C_p to be less than 10^{-7} was adopted. It is known that the gradient ascent scheme, as well as its convergence, is highly sensitive to its initial parameters and inputs. Hence, for future work, it is recommended to introduce some higher fidelity methods of determining step length such as backtracking line search or exact line search if possible.

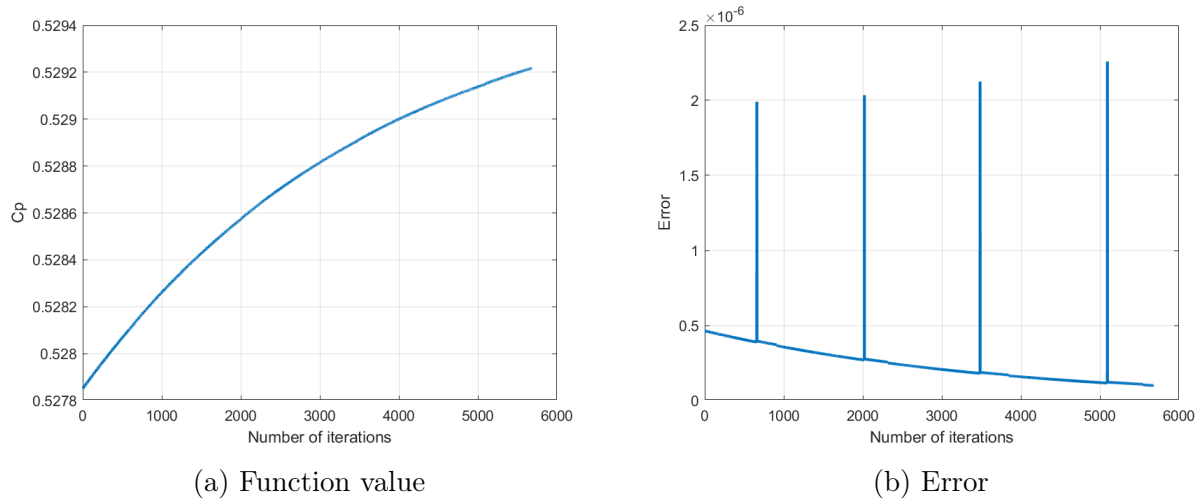


Figure 9: Function value and error as a function of iteration number for maximizing local power coefficient.

Figure 9 presents function value and error as a function of iteration number. One could note that the improvement in the power coefficient is not significant which aligns with the previously stated hypothesis that the blade section has been previously optimized. The results of the optimization process, including local power coefficient with associated chord and angle θ , are summarized in Table 4.

Table 4: Optimized through gradient ascent values of local power coefficient, chord and angle θ .

	Local power coefficient C_p [-]	Chord c [m]	Angle θ [deg]
Optimized values	0.5294	2.22	-2.176

The chord length of 2.22 [m] is significant as it specifies the size of the blade's airfoil and hence aerodynamic characteristics. An airfoil with a shorter chord can contribute to less lift but may also decrease drag. Seeing that the power is limited to 10.64[MW], a shorter chord may be more beneficial by reducing material costs.

List of Figures

1	Process diagram of BEM algorithm used to analyze Wind turbine performance.	1
2	Contour plots using Glauert correction as momentum equation	2
3	Contour plots using Madsen correction as momentum equation	2
4	Rotational speed of wind turbine as a function of wind speed, for operational velocities.	4
5	Blade pitch angle as a function of wind speed.	5
6	Power and thrust curves as a function of wind speed.	5
7	Power and thrust coefficients as a function of wind speed.	6
8	Normal and tangential forces as a function of radius for a range of wind speeds computed using Ashes and BEM algorithm.	8
9	Function value and error as a function of iteration number for maximizing local power coefficient.	10

List of Tables

1	Highest obtainable power coefficient, and the corresponding tip speed ratio and pitch angle values.	3
2	Annual Energy Production for cut-out speeds $V_{out} = 25[\text{m/s}]$ and $V_{out} = 20[\text{m/s}]$	8
3	Initial values of local power coefficient, chord and angle θ	9
4	Optimized through gradient ascent values of local power coefficient, chord and angle θ	10
5	Task Contribution by Group Members	III

Nomenclature

BEM Blade Element Method

VSPR Variable-speed pitch regulation

References

- [1] M. O. L. Hansen, *Aerodynamics of wind turbines*. New York, NY: Routledge, 2015.
- [2] M. O. L. Hansen, “BEM 1 Lecture,” Sept. 2024.
- [3] C. Bak, F. Zahle, R. Bitsche, T. Kim, A. Yde, L. C. Henriksen, A. Natarajan, and M. Hansen, “Description of the dtu 10mw reference wind turbine,” tech. rep., DTU Wind, 2013.

8 Appendix A

46300 - Group Contract

- **Project Manager:** Jakub Kuleta
- **Main Reporter:** Szczepan Letkiewicz
- **Main Coder:** both

Expectations

We collectively expect an above-average performance. The assignments are not our highest priorities, so a grade of 12 is not likely. Somewhere in the range of 7-10 will be satisfactory for our ambitions.

When

Tasks will be assigned to individual group members collectively, after which each member can complete them at their own time and discretion. An interim deadline will be set to 3 days before the final submission so that each group member can look over the report and make any final adjustments if necessary.

How

The assignment will be divided into sections that require collaboration and which can be completed individually. The fundamentals of BEM will be, for instance, a topic that all members of the group are expected to know on their own, whereas some of the difficult niche questions may be answered by certain members.

Amount

Depending on the amount of work, full flexibility of each member is expected.

Flexibility

All members are expected to complete their tasks and attend meetings arranged beforehand. If any spontaneous meetings were to occur, the other members are to be invited but not required to join.

Presence

Members are expected to be physically present at all pre-arranged meetings. This excludes spontaneous meetings.

Repercussions

A 2-strike system will be enforced: after two strikes, the member will be reported to the Professors.

Signed by - Group 25
12/09/2024

Task division

Task	Jakub Kuleta (%)	Szczepan Letkiewicz (%)
Question 1	30%	70%
Question 2	80%	20%
Question 3	100%	0%
Question 4	0%	100%
Question 5	100%	0%
Question 6	100%	0%
Report writing	30%	70%

Table 5: Task Contribution by Group Members